SARCathon 2026 — Stage 2 (Grand Finale)

AXI4-to-PCIe TLP Bridge

Optimize. Implement. Tape Out.

Final technical report — RTL optimization, verification, synthesis and RTL-to-GDSII implementation

Report date: 23 October 2026

Submission deadline: 24 October 2026

Scope statement: The design uses an abstract packet-level TLP interface. It is not a PCIe 6.0 compliant link (no FLIT framing, CRC/FEC, PIPE/PHY, credits or replay), and the result is a hardened macro GDSII, not a fabricated chip.

1. Executive summary

Stage 1 delivered a working AXI4-slave to abstract-PCIe-TLP bridge. Stage 2 took that design through five phases: baseline audit, architecture/performance enhancement, constrained-random verification, synthesis with PPA analysis, and physical design to GDSII.

- **Performance:** 64-beat writes complete 1.83× faster and 64-beat reads 9.4× (8-cycle completion latency) to 13× (32-cycle latency) faster than the Stage 1 baseline, with 4× fewer TLPs (burst coalescing).
- **Verification:** 6,154 / 6,154 self-checking tests and 13,719 assertion evaluations pass with zero failures across 8 random seeds; the synthesized gate-level netlist reproduces RTL behaviour cycle-for-cycle on 3 seeds.
- **Synthesis:** the optimized RTL is 33.1 % smaller than the Phase 2 RTL at its new defaults (78,995 µm² vs 118,121 µm² pre-placement) and 16.1 % smaller at identical parameters.
- **Physical design:** the optimized design completed the LibreLane flow (exit code 0) and produced GDSII with zero detailed-route, Magic and KLayout DRC errors, zero LVS errors and zero XOR differences. Versus the Phase 2 reference run through the same flow: 31.5 % less instance area, 22.5 % less estimated power, 38.8 % less routed wire.
- **Open item, stated plainly:** post-route setup timing is not closed at the 10 ns target (worst slack −1.97 ns, 3,347 violating endpoints) and some max-slew / capacitance / fanout checks still fail. The run used LibreLane's generic fallback SDC. The design is therefore a complete RTL-to-GDSII implementation that is DRC/LVS clean, but not a timing-closed signoff.

1.1 Headline results

Metric	Baseline / reference	Optimized	Change
64-beat write (cycles)	209 (Stage 1)	114	1.83× faster
64-beat read, latency 8 (cycles)	721 (Stage 1)	77	9.4× faster
64-beat read, latency 32 (cycles)	2,257 (Stage 1)	173	13.0× faster
TLPs for 64 beats	64 (Stage 1)	16	4× fewer
Synth area, SKY130 HD (µm²)	118,121 (Phase 2)	78,995	−33.1 %
Post-route instance area (µm²)	323,738	221,869	−31.5 %
Die area (µm²)	343,449	238,198	−30.6 %
Estimated total power (mW)	69.70	54.04	−22.5 %
Hold violations	57	0	eliminated
Setup worst slack @10 ns (ns)	−3.554	−1.969	+1.585 ns

1.2 How Stage 2 was completed: five phases

Stage 2 was too large to treat as a single task, so the work was divided into five sequential phases. Each phase had its own goal, its own evidence (logs, reports, scripts) and a pass criterion, and each one built on the output of the previous phase. This kept every claim in this report traceable to a specific log or report file, and it meant a problem in one phase (for example the Phase 5 tool-environment failure) could be fixed without disturbing the verified results of the earlier ones.

Phase	Goal	What was done	Outcome
1. Baseline audit	Freeze and understand the Stage 1 design	Stage 1 RTL and testbench preserved byte-for-byte with SHA-256 hashes; baseline re-	69 / 69 checks; weaknesses identified

Phase	Goal	What was done	Outcome
		run; audit of throughput, address checks, WLAST handling and completion matching	
2. Architecture and performance	Remove the Stage 1 bottlenecks	Write FIFO, burst coalescing (one TLP per burst), multiple outstanding reads with tags, in-order R return, protocol checks, parameterization	1.83× writes, 9.4–13× reads, 4× fewer TLPs
3. Functional verification	Prove the new design is correct	Constrained-random regression over 8 seeds, handshake and ordering assertions, directed corner cases, packet-processing and ML-inference traffic profiles	6,154 / 6,154 checks; 0 assertion failures
4. Synthesis and PPA	Measure and reduce area and critical paths	Yosys + SKY130 synthesis, seven RTL optimizations, baseline vs optimized comparison, gate-level simulation of the synthesized netlist	−33.1 % area; netlist cycle-identical to RTL
5. Physical design	Implement the design to GDSII	LibreLane flow: floorplan, placement, CTS, routing, STA, DRC/LVS, GDSII, for the optimized and reference RTL	GDSII produced; DRC/LVS clean; timing not closed

The phases map directly onto the Stage 2 tasks: Phases 2 and 4 cover refining the RTL for performance, area and reliability; Phase 3 covers exhaustive verification and the packet-processing and ML workloads; Phase 4 covers synthesis and analysis; and Phase 5 covers OpenROAD/OpenLane physical design and the final GDSII. Phase 1 provided the frozen baseline against which every improvement is measured.

2. Stage 2 requirements and status

The Stage 2 brief asks for: refined RTL (performance, area, reliability); exhaustive functional verification; network packet-processing and ML-inference workloads; synthesis and implementation analysis; physical design with OpenROAD/OpenLane; and the final GDSII. The FPGA-tested AXI4-to-PCIe IP is a bonus item.

Stage 2 task	Status	Evidence
Refine RTL for performance, area, reliability	Done	Phase 2 + Phase 4 changes; Sections 4 and 6
Exhaustive functional verification	Done (simulation)	8-seed regression, 6,154 checks, assertions, directed corner cases; Section 5
Network packet-processing and ML-inference workloads	Done (synthetic profiles)	Dedicated traffic profiles in the Phase 3 testbench; not replayed production traces
Synthesize RTL and analyze results	Done	Yosys + SKY130 HD, baseline vs optimized, gate-level simulation; Section 6
Physical design with OpenROAD/OpenLane	Done — timing not closed	LibreLane 3.0.16 run; Section 7
Generate final GDSII	Done	sarcathon_axi_pcie_bridge.gds and .klayout.gds, top cell sarcathon_axi_pcie_bridge
Bonus: FPGA-tested AXI4-to-PCIe IP	Not done	No FPGA build was performed; listed as future work

3. Design overview

The bridge accepts AXI4 write and read transactions on a slave port and converts them to abstract PCIe memory-write (MemWr) and memory-read (MemRd) TLPs on a transmit interface; completions (CplD) return on a receive interface and are converted into AXI read data.

Feature	Implementation
Buffering	Write-data FIFO (W_FIFO_DEPTH, default 2×MAX_BEATS) decouples W beats from AW and from downstream acceptance; W may arrive before AW.
Burst coalescing	One MemWr TLP per AXI write burst and one MemRd TLP per read burst (payload up to MAX_BEATS beats).
Multiple outstanding reads	TAG_COUNT tags; completions matched by tag and accepted only for sent, not-yet-completed tags.
Ordering	AXI R data returned in AR order even if completions arrive out of order (response-order queue).
Write pipelining	Next AW accepted once the previous write TLP is accepted downstream, without waiting for the B handshake.
Protocol checks	Full-width size, INCR only, length ≤ MAX_BEATS, natural alignment, no 4 KiB crossing, WSTRB and WLAST position. Bad reads return one SLVERR beat; bad writes are drained and answered SLVERR with no TLP.
Parameters	DATA_WIDTH, ADDR_WIDTH, AXI_ID_WIDTH, MAX_BEATS, TAG_COUNT, W_FIFO_DEPTH. Defaults: DATA_WIDTH 64, ADDR_WIDTH 32, ID 4, MAX_BEATS 4, TAG_COUNT 4, W_FIFO_DEPTH 8.

Link-efficiency estimate for coalescing (back-of-envelope, assuming about 24 B of per-TLP overhead): 8 B payload per TLP is about 25 % efficient; 32 B (4 beats of 64 bit) is about 57 %; 128 B (MAX_BEATS = 16) is about 84 %. These are estimates, not measurements, and PCIe 6.0 FLIT-mode overheads differ. Merging address-contiguous bursts across AXI bursts was evaluated and deliberately not implemented, because each burst still needs its own response and error handling.

4. Performance enhancement (Phase 2)

Same workload on every design: 16 bursts × 4 beats, 64-bit data, master and endpoint always ready, read completions returned a fixed number of cycles after the MemRd TLP is accepted. Read data is checked beat by beat; there are 0 data errors in every run.

Scenario	Stage 1	Phase 2 RTL	Final RTL	Final vs Stage 1
64-beat write (cycles)	209	129	114	1.83×
Write beats/cycle	0.306	0.496	0.561	
64-beat read, latency 8 (cycles)	721	77	77	9.4×
64-beat read, latency 32 (cycles)	2,257	173	173	13.0×
TLPs for 64 beats	64	16	16	4× fewer

4.1 Read latency hiding versus TAG_COUNT

TAG_COUNT	Latency 8	Latency 32	Latency 64
1	0.249	0.100	0.056
2	0.481	0.197	0.110
4 (default)	0.831	0.370	0.213

TAG_COUNT	Latency 8	Latency 32	Latency 64
8	0.831	0.587	0.370
16	0.831	0.634	0.481

Values are read beats per cycle. Throughput saturates near 0.83 beats/cycle once TAG_COUNT × beats-per-burst covers the round-trip latency. Each additional tag costs DATA_WIDTH × MAX_BEATS storage flops, so TAG_COUNT is the main area-versus-latency-tolerance knob.

4.2 Known throughput limit

Writes take about 7 cycles per 4-beat burst (ideal 4), because the collection register also sources the TX data and the next burst cannot start collecting until the current TLP is accepted. A ping-pong collection buffer would approach 1 beat/cycle for roughly +6–8 % area; it is documented as the next step and was not implemented.

5. Functional verification (Phases 1 and 3)

5.1 Baseline audit

The Stage 1 RTL and testbench are frozen byte-for-byte with SHA-256 hashes; re-running the baseline reproduces 69 / 69 checks. The audit identified one outstanding transaction, one TLP per beat, no alignment or 4 KiB checks, and completion matching against the AXI ID — which motivated the Phase 2 redesign.

5.2 Constrained-random regression

The self-checking testbench randomizes burst length, payload, AW-versus-W ordering, TX/R/B backpressure (0–3 cycles), completion latency (0–11 cycles) and completion order. A concurrent phase runs six independent threads (AW/W producer, AR producer, TX endpoint, completion endpoint, R consumer, B consumer) with scoreboards for MemWr order, MemRd address/length, R data/ID/LAST in AR order, and B ID in AW order.

Seed	Checks	Assertions (X-on-VALID / W-FIFO / stall)	Result
13579bdf	748 / 748	1,704 (617 / 675 / 412)	PASS
00000001	748 / 748	1,722 (606 / 715 / 401)	PASS
deadbeef	793 / 793	1,725 (634 / 671 / 420)	PASS
2468ace0	758 / 758	1,734 (616 / 709 / 409)	PASS
0badf00d	823 / 823	1,743 (629 / 705 / 409)	PASS
12345678	763 / 763	1,703 (615 / 681 / 407)	PASS
cafef00d	783 / 783	1,733 (633 / 679 / 421)	PASS
7fffffff	738 / 738	1,655 (576 / 706 / 373)	PASS
Total	6,154 / 6,154	13,719 evaluations, 0 failures	PASS

5.3 Directed corner cases

Reset values; partial WSTRB; early WLAST; missing/late WLAST followed by a clean write; FIXED burst; unsupported size; oversized burst; misaligned address; 4 KiB-crossing read and write; tag exhaustion; unknown, duplicate/stale and post-reset completions; completion error mapped to SLVERR; W FIFO full; AW accepted while B is pending; and reset during TX stall, outstanding read, partial write collection, B stall,

and mid-burst R stall — each followed by a recovery transaction. The earlier Phase 2 directed testbench also passes on the final RTL (66 / 66).

5.4 Application workloads

- **Network packet processing:** many small sequential writes modelling a packet stream into host memory.
- **ML inference:** batches of 4-beat feature/weight reads with all tags outstanding and completions returned in reverse order.

These are synthetic profiles; they are not replayed traces from a real SmartNIC or accelerator.

5.5 Functional coverage and limits

Every scenario bin is hit on every seed (for example 48 mixed concurrent bursts, 24 AW-while-B-pending events, 20 tag-reuse read bursts, 7 out-of-order completions on the default seed, 5 reset-during-activity cases). These are scenario counters, not tool-computed coverage. There is no formal proof and no code coverage. A bug found in the testbench itself during this phase (a shared temporary variable causing a deadlock) was fixed and documented. Extra W beats beyond AWLEN+1 from a non-compliant master are not recovered.

6. Synthesis and PPA (Phase 4)

Yosys with ABC, SKY130 sky130_fd_sc_hd typical corner (tt_025C_1v80), 10 ns clock, identical script for every row. Area is pre-placement standard-cell area (no taps, fill, CTS or resizer buffers).

Configuration	Area (µm²)	vs Phase 2	Cells	Sequential	Leakage (nW)
Stage 1 baseline	15,038	−87.3 %	1,347	356	6.5
Phase 2 RTL	118,121	—	9,275	3,088	46.6
Final RTL, FIFO depth 16	99,061	−16.1 %	5,649	2,655	32.7
Final RTL, defaults (depth 8)	78,995	−33.1 %	5,071	2,124	26.3

The Stage 1 baseline is smaller only because it has no buffering, one outstanding transaction and a 64-bit data path. The like-for-like comparison is Phase 2 versus the final RTL.

6.1 RTL optimizations

#	Change	Effect
1	tx_tlp_data driven from the write-collection register; duplicate packet registers removed	−256+ flops; removes the 299-fanout TX-launch net
2	W FIFO stores a one-bit "strobe all ones" flag instead of full WSTRB	−7 bits per FIFO entry
3	Pure datapath registers have no reset; every control/VALID flop keeps its async reset	Smaller flops; X-safety proven by X-on-VALID assertions in RTL and gate-level simulation
4	Rejected reads return zero by masking s_axi_rdata	Removes a 1,024-bit clear path
5	Default W_FIFO_DEPTH 16 → 2×MAX_BEATS (8)	−536 flops
6	Critical-path fix: w_final subtract-and-compare replaced by a registered flag	Critical path moved off the write collector
7	Write pipelining (accept next AW after TLP accepted)	+13 % write throughput

6.2 Post-synthesis functional check

The Phase 3 testbench was run against the synthesized SKY130 netlist (DATA_WIDTH 32, MAX_BEATS 4, TAG_COUNT 4, W_FIFO_DEPTH 8) on three seeds: 748/748, 748/748 and 793/793 checks pass, finishing at exactly the same simulation time as RTL with identical assertion counts. This is zero-delay simulation without SDF back-annotation; timing is covered by static timing analysis in Section 7.

Parameter elaboration passes for 9 configurations (DATA_WIDTH 32/64/128, MAX_BEATS and TAG_COUNT 1…16 including non-powers of two, W_FIFO_DEPTH 2…32, 64-bit addresses), and Verilator -Wall lint is clean on the same sets.

7. Physical design (Phase 5)

7.1 Flow and setup

Item	Value
Flow	LibreLane 3.0.16, Classic flow
Tools	Yosys 0.62, OpenROAD (dcf36133…), Magic 8.3.623, KLayout 0.30.7
PDK / cells	sky130A (ciel 8afc8346…), sky130_fd_sc_hd
Clock	clk, 10 ns (100 MHz target)
Synthesis strategy	AREA 0
Floorplan	Relative sizing, 40 % target core utilization, aspect ratio 1
I/O plan	Custom pin_order.cfg: TX TLP north, RX completion south, AXI write channels west, AXI read channels + clock/reset east (835 signal pins)
Timing repair	Post-global-route resizer timing repair enabled
Runs	Optimized RTL, and frozen Phase 2 RTL as reference, identical constraints
Environment	Google Colab with Nix-installed tools (a Python _opcode import problem in the Yosys helper was diagnosed and fixed before the successful run)

7.2 Results: optimized versus reference

Metric	Reference (Phase 2)	Optimized	Change
LibreLane exit code	2	0	
Flow run time (min)	125.5	66.5	
Instances	56,023	38,619	−31.1 %
Instance area (µm²)	323,738	221,869	−31.5 %
Die area (µm²)	343,449	238,198	−30.6 %
GDS size (µm)	580.71 × 591.43	482.73 × 493.44	
Utilization	62.93 %	62.83 %	
Setup worst slack (ns)	−3.554	−1.969	+1.585
Setup TNS (ns)	−1182.322	−1014.689	
Setup violations	3,928	3,347	−14.8 %
Hold worst slack (ns)	−0.234	+0.108	
Hold violations	57	0	eliminated

Metric	Reference (Phase 2)	Optimized	Change
Total power (W)	0.06970	0.05404	−22.5 %
Internal power (W)	0.03962	0.03197	−19.3 %
Switching power (W)	0.03007	0.02207	−26.6 %
Leakage power (W)	3.59e-7	2.42e-7	−32.7 %
Routed wirelength (µm)	686,801	420,175	−38.8 %
Max-slew violations	3,705	1,704	reduced
Max-capacitance violations	7	1	reduced
Max-fanout violations	117	61	reduced

7.3 Signoff checks — optimized design

Check	Result	Comment
Detailed-route DRC	0	Pass
Magic DRC	0	Pass
KLayout DRC	0	Pass
LVS errors	0	Pass
XOR differences (Magic vs KLayout GDS)	0	Pass
Illegal overlaps / disconnected pins / unmapped cells	0 / 0 / 0	Pass
Power-grid violations / lint errors	0 / 0	Pass
Hold timing	+0.108 ns, 0 violations	Pass
Setup timing @ 10 ns	−1.969 ns, 3,347 violations	FAIL — not closed
Max slew / capacitance / fanout	1,704 / 1 / 61	FAIL — not closed

The GDS top cell is sarcathon_axi_pcie_bridge with 42 layers in the main GDS. Both a Magic stream-out (.gds) and a KLayout stream-out (.klayout.gds) were produced for each run.

7.4 Interpreting the timing result

- The reported Fmax estimate from the worst setup slack is about 83.6 MHz (1000 / (10 + 1.969)), against the 100 MHz target.
- PNR_SDC_FILE and SIGNOFF_SDC_FILE were undefined, so LibreLane applied its generic fallback constraints, including on a design with 835 I/O pins. Timing against the intended SoC I/O budget has not been analysed.
- Phase 4 ABC estimates put internal logic depth at about 2.8 ns, so the failing paths are likely dominated by wire load, high-fanout nets and I/O constraints rather than pure logic depth. This has not been confirmed from the path reports.
- The reference run exit code 2 is caused by deferred hold violations in the max_ss_100C_1v60 and nom_ss_100C_1v60 corners; it is a baseline for comparison, not a signoff run.
- Phase 4 area (78,995 µm²) is pre-placement cell area; Phase 5 instance area (221,869 µm²) is a post-route figure that includes tap cells, fill, clock-tree and timing-repair buffers. The two are not directly comparable, and the breakdown should be confirmed from metrics.json.
- Power is an OpenROAD estimate using default switching activity, not a measurement.

8. Limitations and honest scope

- Abstract packet-level TLP interface: no PCIe 6.0 FLIT framing, CRC/FEC, PIPE/PHY, credits or replay. No PCIe 6.0 compliance is claimed.
- Full-width, aligned INCR bursts up to MAX_BEATS with all-ones write strobes; one write burst collected at a time; one abstract completion per burst (no completion splitting).
- Verification is simulation-only; no formal proof, no code coverage; workload profiles are synthetic.
- Physical design targets a hardened macro, not a pad-framed chip or multi-project-wafer submission; it has not been taped out.
- Post-route setup timing, max-slew, max-capacitance and max-fanout are not closed at 10 ns (Section 7).
- The bonus FPGA-tested IP was not built.

9. Reproducing the results

9.1 Phases 1–4 (local)

Requires Icarus Verilog; Verilator, Yosys and Python 3 are used when present (YosysHQ OSS CAD Suite recommended). From the repository root:

```
bash run_all.sh        # Linux/macOS
.\run_all.ps1         # Windows PowerShell
```

9.2 Phase 5 (Google Colab)

Upload phase5_physical_design/SARCathon_Phase5_LibreLane.ipynb to Colab and run all cells. The notebook embeds the RTL, configurations and pin order with SHA-256 hashes, installs LibreLane via Nix, runs the optimized and reference flows, and exports a results archive containing GDSII, reports and a summary.

9.3 Repository layout

Path	Contents
rtl/	Current design: sarcathon_axi_pcie_bridge.v
phase1_baseline_audit/	Frozen Stage 1 RTL, testbench, report, manifest, audit
phase2_architecture/	Architecture notes, directed and throughput testbenches, logs
phase3_verification/	Phase 3 testbench, regression logs for 8 seeds
phase4_synthesis/	Synthesis scripts, reports, frozen Phase 2 reference RTL, gate-level netlist and simulation logs
phase5_physical_design/	LibreLane configs, pin order, Colab notebook, results/ (GDSII, reports, summary)
README.md, PHASE_STATUS.md, SOURCE_MANIFEST.md	Entry point, per-requirement status, file hashes

10. Future work

- **Close timing:** supply a real SDC (I/O delays, clock uncertainty) and re-run at a relaxed period (for example 12–15 ns), or add AR/AW register slices and a one-hot registered head-tag pointer to cut the 266-load read-mux net.
- **Write throughput:** ping-pong collection buffer to approach 1 beat/cycle on writes.

- **FPGA bonus:** map the bridge onto an FPGA with a test harness to earn the hardware-validated IP bonus.
- **Verification depth:** add SVA/formal checks on handshake and ordering properties and measure code coverage.